

# ECON 311

## Lecture 14: The IS-MP Model, part III

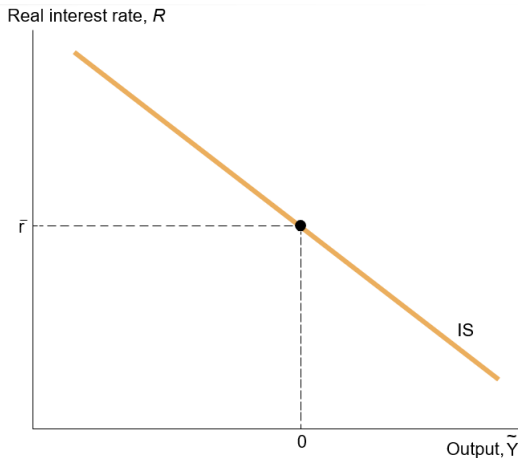
Goods Market

Monetary Policy

The Phillips Curve

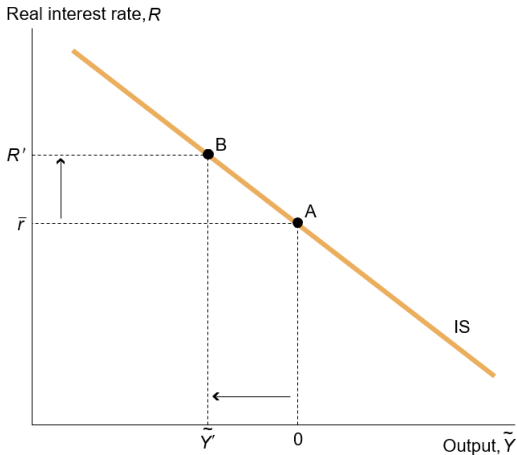
## IS-curve with new scale

$$\text{IS-curve: } \tilde{Y}_t = \left[ \frac{1}{1-b_c} \right] * \left[ \bar{a} - \bar{b}_i(R_t - \bar{r}) \right] - 1, \quad \bar{a} \left[ \frac{1}{1-b_c} \right] = 1$$



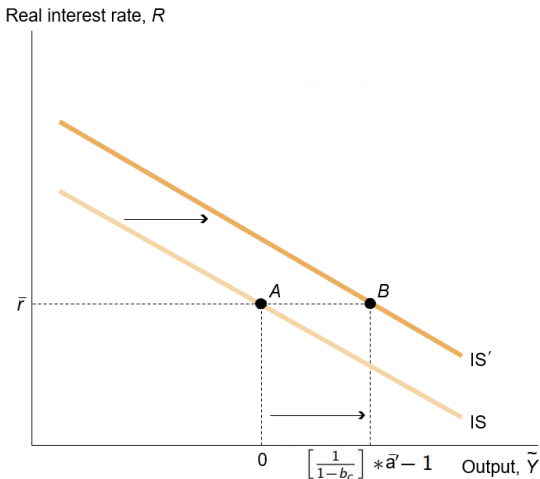
## Changes in the real interest rate

$$\text{IS-curve: } \tilde{Y}_t = \left[ \frac{1}{1-b_c} \right] * \left[ \bar{a} - \bar{b}_i(R_t - \bar{r}) \right] - 1, \quad \bar{a} \left[ \frac{1}{1-b_c} \right] = 1$$



## Temporary Aggregate Demand Shock

$$\text{IS-curve: } \tilde{Y}_t = \left[ \frac{1}{1-b_c} \right] * [\bar{a} - \bar{b}_i(R_t - \bar{r})] - 1, \quad \bar{a} \left[ \frac{1}{1-b_c} \right] = 1, \quad \bar{a}' > (1 - b_c)$$

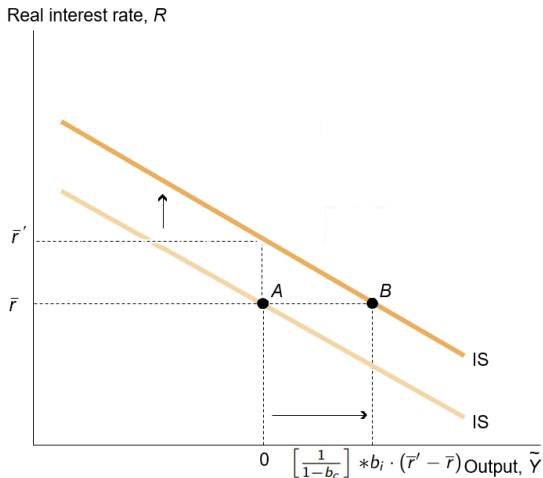


# Permanent Aggregate Shock

- Permanent shocks impact potential output and originate from the production side of the economy, which have two effects:
  1. Permanent shocks affect demand parameters, which affect both long- and short-run output levels potentially equally.
  2. Permanent production shocks affect the *MPK*
    - ▶ This has an effect in the long-run because  $\bar{r} = MPK$
    - ▶ This results in output overshooting in the SR as the economy adjusts to its new potential output.

## E.g. Permanent TFP shock

$$\text{IS-curve: } \tilde{Y}_t = \left[ \frac{1}{1-b_c} \right] * [\bar{a} - \bar{b}_i(R_t - \bar{r})] - 1, \quad \bar{a} \left[ \frac{1}{1-b_c} \right] = 1, \quad \bar{r}' > \bar{r}$$



Goods Market

Monetary Policy

The Phillips Curve

# Monetary Policy

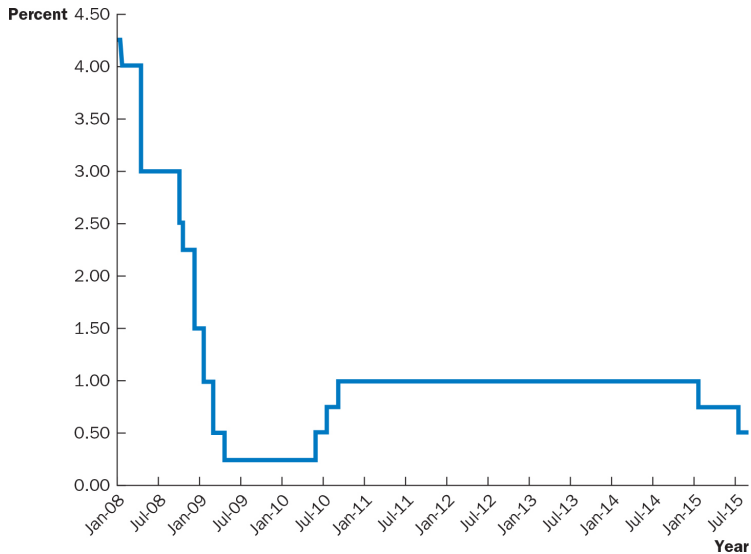
- The BoC currently has three main monetary policy variables at its disposal.
  1. Overnight rate
  2. Open-market operations
  3. Reserve requirements

# Overnight rate

- Overnight rate: The interest rate on very short-term loans between commercial banks.
  - ▶ The BoC can indirectly change the interest rate at which commercial banks lend to each other.
- When banks lend too much and do not meet the reserve requirement, they are able to borrow short-term from other commercial banks which have excess reserves or from the BoC itself at the overnight rate to meet the requirement.
  - ▶ A high overnight rate encourage banks to lend less.
- The BoC announces changes in the overnight rate on eight fixed days each year (around every month and a half).

# Overnight rate

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## From nominal to real interest rate

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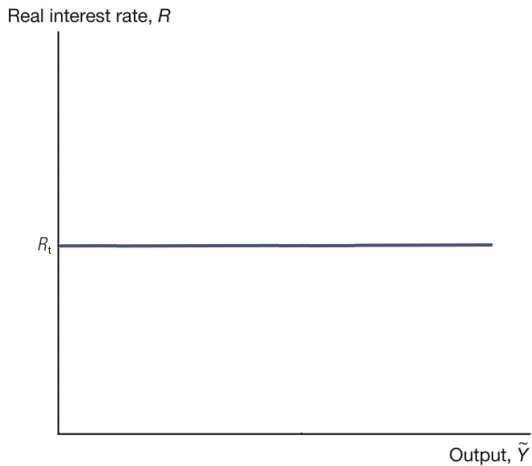
- By modifying the overnight rate, the BoC affects the nominal interest rate in the market.
  - ▶ Commercial banks' additional borrowing costs is pass on to borrowers
- We can study the effect of changes in the nominal interest rate on the real interest rate using the fisher equation:

$$R_t = i_t - \pi_t$$

- ▶ We are going to assume that in the SR, prices are sticky and only change slowly over time.
- ▶ In the SR,  $\Delta\pi$  do not completely offset  $\Delta i$  (this is different in the long-run).
- ▶ Therefore, in the SR, CB monetary policy has an indirect influence on the real rate return,  $R$ .

# The MP curve

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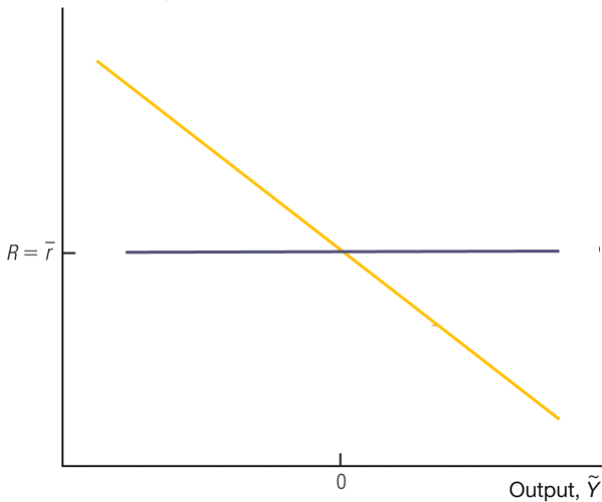


# The IS-MP model

- The MP curve
  - ▶ Illustrates the central bank's ability to set the real interest rate at a particular value (horizontal line)
- The IS curve
  - ▶ Illustrates the negative relationship between short-run output and interest rates (through changes in aggregate expenditure)

# The IS-MP model

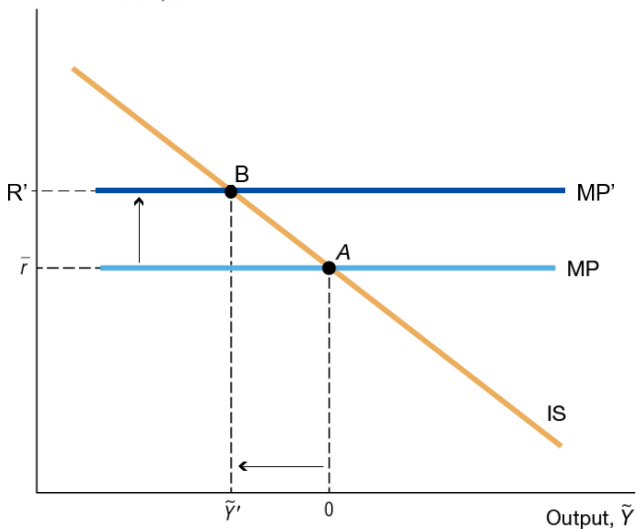
The MP Curve in the IS-MP Diagram  
Real interest rate,  $R$



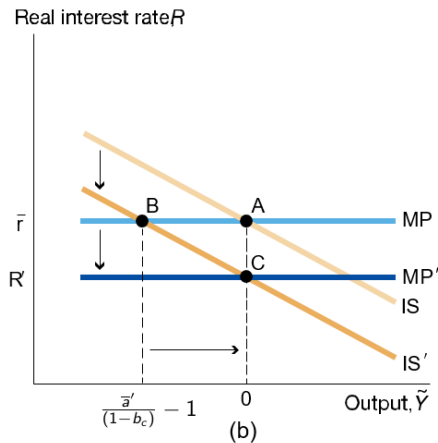
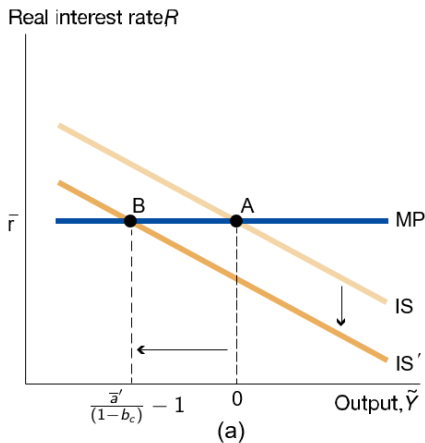
# The IS-MP model

## Raising the Interest Rate in the IS-MP Diagram

Real interest rate,  $R$



# Temporary Negative Demand Shock



# IS-MP Model

- In this simple model, the CB can avoid recessions or overheating economies through active monetary policy.
  - ▶ However, in practice there is usually a lagged between shocks and the response of the CB due to two factors:
    1. It usually takes time for policy makers to understand the extent of the shocks.
    2. It takes time for the policy to affect the economy (between 6-18 months for changes in the interest rate to have substantial effect on economy activity).
  - ▶ Business Cycles are the result of the interaction of shocks and the response of the CB and governments to these shocks.

Goods Market

Monetary Policy

The Phillips Curve

# The Phillips Curve

- Inflation is defined as:

$$\pi_t = \frac{P_{t+1} - P_t}{P_t}$$

- Firms set their prices on the basis of:
  - ▶ Their expectations of the economy-wide inflation rate
  - ▶ The state of demand for their product

# The Phillips Curve

- Inflation is then determined by the following equation:

$$\pi_t = \pi_t^e + \bar{v}\tilde{Y}_t$$

- We are going to assume:  $\pi_t^e = \pi_{t-1}$ 
  - ▶ Firms adjust their forecast slowly.
  - ▶ This embodies the sticky inflation assumption

# The Phillips Curve

- Using the equations:

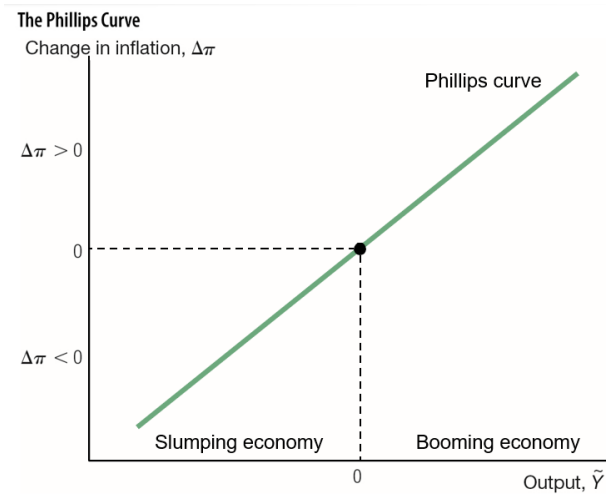
$$\pi_t = \pi_{t-1} + \bar{v}\tilde{Y}_t$$

$$\Delta\pi_t = \pi_t - \pi_{t-1}$$

- Therefore, the Phillips curve can be expressed as:

$$\Delta\pi_t = \bar{v}\tilde{Y}_t$$

# The Phillips Curve

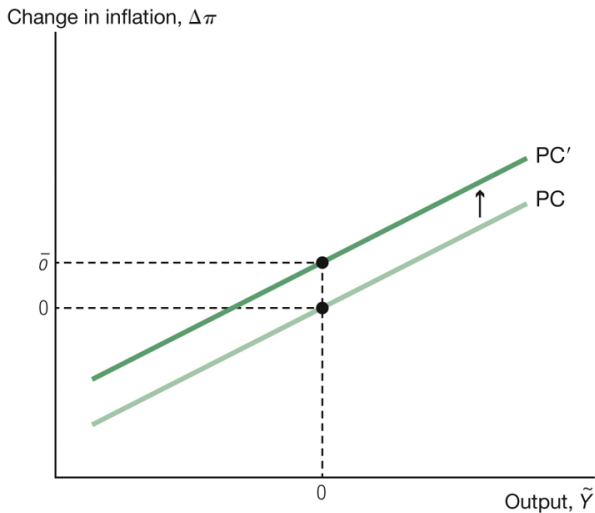


# Inflation Shocks

- In our IS-MP model, we are going to also include inflation shocks (called cost-push inflation)
  - ▶ We are going to include these shocks in our Phillips curve as follows:

$$\Delta\pi_t = \bar{v}\tilde{Y}_t + \bar{o}$$

# The Phillips Curve with cost-push inflation



# Sticky Inflation

- The sticky inflation assumption is a very important property of this model
  - ▶ Inflation displays inertia so any change to prices in the present affects inflation in the future
  - ▶ Shocks to the Phillips Curve have long memory
  - ▶ A one time economic policy affects inflation rates in multiple periods